Certified But Imperfect: Investigating The Role of AI Certifications And System Performance on Trust in And Reliance on AI Systems

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Abstract

While regulatory frameworks call for the implementation of AI certifications, empirical knowledge about how such certifications affect interactions is still scarce. In this work, we examined how AI certifications affect users' trust and reliance. In addition, we examined whether certifications elevate user expectations and whether unmet expectations subsequently reduce trust. In a 2 (certification vs no certification) x 2 (reliability: high vs low) between-subjects online study, N = 644 participants had to identify bacterial infestation in pictures with the help of an AI. Our results show that, before interacting with the AI, participants trusted the certified system more and showed reduced vigilance. However, these effects disappeared post-interaction, where, instead of the certification, system reliability significantly affected trust and vigilance. Notably, certifications did not raise expectations per se, but instead amplified the impact of system reliability on user trust. Additional exploratory results showed that the certification supported appropriate reliance.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

Keywords

AI certification, trust, reliance, expectation violation

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1 Introduction

Artificial intelligence (AI) systems are becoming increasingly integrated into everyday life, from mundane everyday tasks to high-risk cases, spanning political, economic, and social domains [9, 77, 90]. Simultaneously, these systems are becoming increasingly automated, complex, and opaque, while also increasing in interconnectivity [68].

As such, AI systems promise significant benefits like improved efficiency, personalization, predictive accuracy, and operational scalability [2, 40, 72]. Yet, their transformative potential comes with growing concerns about unintended and harmful consequences, such as generating realistic deepfakes, disrupting educational integrity, reproducing and amplifying societal biases, and causing discriminatory or otherwise opaque decision-making processes [85, 97, 106, 108]. Their rapid and far-reaching proliferation, particularly with the introduction of generative AI, accentuates the need for regulatory frameworks to foster appropriate trust in these systems and enforce compliance with human rights [22].

Recognizing this need, several initiatives have introduced regulatory frameworks to establish standards for trustworthy AI, signaling a shift toward formalized AI auditing and certification to ensure the safety, fairness, and accountability of these systems. For example, the European Union's AI Act (EU AIA) aims to "promote the uptake of human-centric and trustworthy artificial intelligence" [33]. These initiatives reflect what Knowles and Richards [56] describe as "trust in AI-as-an-institution" (p. 265) - trust grounded not in individual system behavior but in a regulatory ecosystem that signals systemic trustworthiness.

Amid these approaches, one recurring proposal across many governance frameworks is the call for AI certifications. Certification schemes are envisioned as tools to assess and communicate compliance with ethical and technical standards, especially in high-stakes contexts. As such, certifications have long played an important role in signaling safety and quality across various domains, including food, pharmaceuticals, aviation, and electrical engineering [28, 51, 58, 75], helping to build public trust and enforce industry standards.

Yet despite their growing policy prominence, there is limited empirical understanding of how AI certification mechanisms will function in practice, especially from the perspective of users. We argue that HCI research has a critical role to play in bridging the disconnect between regulatory ambitions and user experience. In this work, we aim to inform the implementation of certifications as user-centered AI governance tools that are not only technically robust but also socially and experientially meaningful. Moreover, prior work in HCI shows that users may misinterpret signals of trustworthiness [55], we focus on how certifications shape expectations toward the system and the interplay of these expectations when interacting with the system. To do so, we are the first study to assess the effect of certifications on trust and reliance before and after interacting with a system, going beyond the scope of previous vignette studies. The two central research questions of our work are:

- RQ1: How do AI certifications shape users' pre and post-interaction perceptions of the system?
- RQ2: How do certifications in combination with the system's performance shape users' trust and reliance?

To answer these questions, we investigated how AI certifications in combination with the system's reliability shaped users' trust and reliance on a fictional AI-powered decision support system. Our findings suggest that, while certifications could raise initial trust levels, user may have over-generalized certifications, extending beyond the scope of what the certification actually covers.

2 Related work

2.1 Current AI regulation and the role of HCI

As AI-enabled systems become increasingly integrated into everyday decision-making contexts, users must navigate complex judgments about when and how much to rely on algorithmic recommendations [103]. This challenge of appropriate reliance becomes particularly acute given that AI failures can result in severe, sometimes fatal, consequences [42]. Especially such high-stakes failures have intensified calls for responsible AI deployment, positioning regulatory frameworks as a primary mechanism for ensuring algorithmic accountability.

Similar to the evolution of AI systems, the regulatory frameworks have undergone a significant change, shifting from voluntary guidelines and soft law approaches toward binding regulatory frameworks [1]. Prominent examples are the European Union's AI Act, the United States' Executive Order 14110, Canada's AI and Data Act, China's Interim Measures for Generative AI Services, and Brazil's AI Bill 2338/2023. The result is a complex, divergent, and fragmented global AI landscape [1]. Moreover, while these regulatory initiatives aim to address AI accountability, individual frameworks face significant implementation challenges. The EU AI Act, for example, has been criticized for ambiguity and for concentrating too much power in corporate players [62], highlighting the gap between regulatory intent and practical effectiveness.

From an HCI perspective, we understand that the effectiveness of these regulatory frameworks ultimately depends on how they translate into user experiences and decision-making contexts. Historically, HCI as an academic discipline has played a crucial role in engaging with and shaping not only technological development itself (see, e.g., [84]) but also societal issues regarding technology, such as the regulation of technologies [43]. As Hochheiser and Lazar [43] have highlighted in their framework for engagement, HCI scholars have (and should) contribute to issues such as, for example, e-health, education, electronic government, privacy, and voting.

Present examples of how the HCI community contributes to the regulation of AI address not only the design and development of solutions to regulatory mandates, but also question and critique the implementation of existing regulations from a human-centric perspective. Most prominently, the EU AIA, for example, demands explainability as a crucial requirement for regulatory compliance under the EU AIA, particularly for high-risk systems. However, existing HCI research indicates that explainability might come with undesired outcomes. For example, Eiband et al. [31] could show that placebic explanations, explanations without conveying information, increased users' trust. Similarly, Ferguson et al. [35] showed that explanations increased users' cognitive load, while Bertrand et al. [6] showcase how cognitive biases affect the accurate processing of explanations.

Besides these empirical results, Urquhart et al. [95] suggest various "legal provocations" (p. 1), aiming to create pathways for the HCI community to contribute to the implementation of the EU AIA. As such, Urquhart et al. [95] argue that the requirements of the EU AIA invite contributions from HCI scholars by, for example, designing and developing appropriate solutions to data quality requirements, risk management, technical documentation, and human oversight. Likewise, Valdez et al. [96] argue that the HCI community plays a crucial role in shaping AI regulation by bridging the gap between technical capabilities and human-centered requirements [96].

Also addressing the implementation of the EU AIA, Sterz et al. [87] first propose an approach for effective human oversight that they, subsequently, juxtapose with the requirements of the EU AIA. Likewise, [60] suggest signal detection theory to better understand how to arrive at effective human oversight. To that end, Laux and Ruschemeier [61] question how the EU AIA's mandate for awareness of the automation bias can be legally enforced. Their analysis suggests that the EU AIA lacks explicit design and context requirements to counter causes of automation bias. Laux and Ruschemeier [61] conclude that, to implement effective safeguards against automation bias, further human-AI interaction is needed. This body of work demonstrates HCI's capacity to critically examine the human implications of regulatory requirements.

In this work, we are particularly interested in the role of certifications and their accompanying certification marks. Being central to many of the regulatory initiatives and auditing strategies, certifications (see EU AI Act or Canada's The Artificial Intelligence and Data Act) aim to signal the trustworthiness of a system. However, how such certifications shape user experiences is less clear. In the next section, we, first, review existing approaches to simplify and communicate AI capabilities to lay users with a special emphasis on visualizations, summarize empirical findings about AI certifications, and point to gaps in the literature.

2.2 Driving attention and signaling importance: Effects of icons, marks, and labels when interacting with AI systems

Different visualization techniques and strategies have been tested to simplify and communicate AI system characteristics, their associated risks and benefits, as well as the overall system trustworthiness to lay users. For example, for the context of AI model cards, shorter text has been shown to increase understandability and interpretability [12]. Likewise, color, especially red, and prominent typography have been associated with increased risk perception and memorability [16, 104, 107] (see [37] for an overview).

Similarly, charts (e.g., [78]), metaphors (e.g., [21]), and icons (e.g., privacy nutrition labels [50], and AI legibility marks Lindley et al. [64]) have been found to support lay user comprehension of complex data environments and AI systems by driving attention and signaling importance. Furthermore, more complex visualizations such as the Impact Assessment Card [11] or AI Failure Cards, [92] have been empirically tested, suggesting that these visualizations successfully communicate the risks and benefits of specific AI systems and help users to find mitigation strategies.

A similar line of research explores the application of labels. For example, analogue to nutrition and energy labels, different works have suggested labels for AI systems, such as the AI Nutrition Facts [47], Open Ethic's Nutrition Label [32], and energy labels [36]. These initiatives oftentimes aim to create comprehensible overviews of certain system characteristics, such as power draw per inference, and computed robustness, but also help users to assess trade-offs such as transparency and accuracy. To that end, insights from Caven et al. [16] show that the effectiveness of such labels is likely driven by users' demand. In their work on the U.S. Cyber Trust Mark, the authors found that, when purchasing a smart device, users were most concerned with the product pricing and least concerned with cybersecurity. Hence, Caven et al. [16] only found an effect of the Trust Mark for users who were generally security aware. Also studying the implementation of the U.S. Cyber Trust Mark, Chen et al. [20] examined how label complexity, different label attributions, and a brief educational intervention could possibly affect users' IoT purchase selections. The authors found that users preferred higher-complexity labels that also informed about data collected, utilized, and shared by the device. The educational intervention improved general understanding of the Trust Mark but did not affect behavior, that is QR code scans, much.

Beyond purchase behavior, labels have also been employed to inform about AI legibility [39, 64, 66]. For example, Gamage et al. [39] tested warning labels to assist users in identifying AI-generated content on social media platforms. Their results show that, while the label had a significant effect on what users perceived to be AI-generated, the label did not affect engagement behavior. Moreover, Gamage et al. [39] could show that trust in the label was related to the design choices and content type.

Building on these results, that is the successful application of icons, marks, and labels, AI certifications, as demanded by various regulatory frameworks, should potentially aid users when interacting with AI systems. However, as we review in the next section, first empirical findings on the effectiveness of AI certifications are mixed.

2.3 AI certifications as signals for trustworthy AI

Certifications generally function "as a crucial mechanism for providing reliable documentation that attests to the fulfillment of predetermined standards" [13, p. 4224], aiming to reduce complexity and uncertainties about products and systems. Once a certification has been issued by an auditing instance (either an organization or an individual), the certification is communicated through a certification mark or seal. As such, they signal to users that a certain service or product fulfills a predefined set of quality marks often related to safety but also origin and manufacturing.

For the context of AI, while various initiatives have introduced certification procedures for AI (see e.g., [3, 24, 38, 100, 112]), up to date, it is still not clear what exactly a certification process should contain, how the process of certifying should look like, and who should be the certifier. To that end, first studies used existing certification frameworks to assess their effectiveness. For example, Autischer et al. [5] documented the practical application and limitations of existing certification catalogues in the light of the AI Act. They identified some limitations of the certification catalogues, such as the need for an active deployment team. Likewise, Brogle et al. [13] propose that AI certifications are an opportunity to "address short-term gaps in regulation" (p. 4227). Through a case study in the financial industry, Brogle et al. [13] provide an analysis of the Responsible Artificial Intelligence (RAI) certification conducted by the Responsible Artificial Intelligence Institute (RAII). The authors conclude that certification-seeking institutions need "actionable insights on what to improve regarding the development, deployment, and maintenance of the certified system" (p. 4235), while "the audit criteria were too complex for digestibility" (p. 4235). Regarding the certifying institution, Blösser and Weihrauch [10] examined who should issue a certification. In their work, the authors found that, over different decision domains, users prefer governmental institutions and NGOs as certifier.

While there are no binding certifications, and it is unclear how a certification process would look like, outside the context of AI, empirical evidence supports the effectiveness of such certifications. In the context of online retail, for example, assurance services such as certifications or seals of approval have been shown to increase consumers' trust, outperforming alternatives such as rating systems that are linked to ambiguity, negativity, and risk [15]. Likewise, Web Assurance Seal Services (WASS) have been shown effective instruments to increase users' trust and mitigate user concerns about e-commerce platforms [52]. Moreover, certification labels were found to reduce perceptions of risk and increasing brand reputation which both helped to increase consumers' trust [93].

Work on these certifications typically evaluate systems with well-understood, relatively stable behavior and clearly defined failure modes. However, AI presents fundamentally different challenges. Unlike traditional systems, AI models often operate as "black boxes," learning complex patterns from data in ways that are not always interpretable or predictable. Their performance can degrade silently over time, be highly sensitive to context, or encode hidden biases from training data [68]. Moreover, AI systems can evolve post-deployment through continuous learning or dynamic interactions with users, making static certification less straightforward.

Adding to these technical challenges, from a user's side, first empirical studies examining AI certifications specifically have shown mixed results for their effectiveness. Although for professional contexts labeling approaches for AI are not new, for example, in the form of data or model cards [25, 76], AI labels for the general public have been less explored [36]. Using an established trust mark, Scharowski et al. [81] found, for example, that certifications, indicated through labels, significantly increased users' trust and willingness to use AI, especially for high-risk contexts. Likewise, de Almeida et al. [27] found that AI certifications increased users' trust and acceptance of AI. Additionally, the authors found that transparent communication about the certification criteria was important.

Contradicting these results, Wischnewski et al. [102] found no effect of an AI trust label. The authors explain their null results that the initial level of users' trust was already high and could not be increased through the presence of an AI certification, indicating a ceiling effect. Similarly, comparing the separate and joint effects of XAI and AI certifications, Szymczyk et al. [91] found no effect of an AI certification on trust, perceived, and actual understanding. The authors suggest that their null findings are either the result of their relatively low-risk context or the use of an unestablished certification mark.

2.4 Certified but still imperfect: On the role of expectancy violations

While previous findings of AI certifications either indicate null effects or positive effects on trust, there is some evidence that certifications might also have detrimental effects. Some of the previous work on certifications could show that users over-relied on and misinterpreted assurances, allowing well-intended trust signals to evoke wrong expectations Kimery and McCord [54], Kirlappos et al. [55], Turow [94]. Coining the *expectation gap*, findings by Houston and Taylor [45], for example, indicated that a WebTrustSM assurance did not communicate the intended purpose (assurance of business and security practices) but instead raised consumers' product quality expectations. Likewise, Lins et al. [65] found that, not meeting expectations due to misattributions of a trust signal's functionality led to increased skepticism and mistrust, undermining the original intended purpose of the signal.

Central to these findings is the role of expectations. As such, expectations play a crucial role in shaping how users interact with systems. Users' preconceptions about a system's functionality, ease of use, or reliability, for example, influence their engagement and emotional responses. Likewise, early works on trust suggest that trust is harmed when expectations are violated and strengthened when met Lewicki et al. [63]. Setting the right expectations is, hence, key to any human-technology interaction [67] and can be shaped by design choices [57].

Various theoretical models explore the role of met and unmet expectations when interacting with technologies, such as the Expectation-Confirmation Model [7], the Expectation Disconfirmation Theory (EDT) [8] or the Unified Theory of Acceptance and Use of Technology (UTAUT) [98]. These models prominently outlined the role of different expectations regarding, for example, the performance of a system or the expected effort to learn how to engage with a system in driving behavioral intentions and user behavior.

For our purpose, we will apply Burgoon's [14] Theory of Expectation Violations which offers insights into how deviations from expected behaviors influence information processing and user perceptions. Specifically, Burgoon [14] proposed that when behavior significantly diverges from expectations, attention shifts from the original subject to the behavioral deviation itself. These violations are subsequently assigned a valence, which is a positive or negative evaluation based on the comparison between the expected and actual behavior.

While Burgoon's [14] theory was originally developed to explain interpersonal communication, its principles can be extended to human-computer interactions. Drawing on Nass et al. [70], who argued that computer-mediated communication is inherently social, the framework of expectation violations can help to understand how users' expectations of AI systems shape their experiences and responses during interaction. This theoretical lens suggests that AI systems, like human interlocutors, are subject to evaluation based on how well they meet or violate user expectations, influencing both user satisfaction and trust in technology.

Further supporting the importance of met and unmet expectations comes from literature discussing the perfect automation scheme. In their influential work,Dzindolet et al. [30] described that some users tend to view automated systems as infallible, holding high performance expectations and engaging in all-or-none thinking about automation. Individuals with this schema expect automated aids to work perfectly, and when errors occur, their trust drops sharply, often leading to abrupt rejection of the system rather than incremental adjustment. However, despite the continuous improvement of AI, errors in predictions are inevitable [19]. Even with a nearly perfect AI system, errors will occur and might induce expectation violations. In this work, we suspect that certifications might raise additional system expectations which, as no system is perfect, might unintended reduce trust more when a system is certified than when it is not.

2.5 Research Gap & Hypotheses

Reviewing the literature on AI certifications, we identified several gaps. For example, all above mentioned empirical studies employed vignettes, measuring perceptions and intentions but not behavior. While perceptions and intentions can be insightful, users did not interact with a system. However, from previous work, we know that system behavior is one of the key predictors of users' trust in AI [49]. To address this gap, in a first step, we differentiated between trust before and after an interaction with a system. To that end, we hypothesized:

H1: Prior to interacting with an AI system, a certified AI system is trusted more than an uncertified AI system.

We assume that this effect is consistent over time, and will also be present after a user has interacted with a system. Moreover, as trust is a predictor for reliance, we also assumed that reliance behavior will be affected:

H2: After interacting with an AI system, a certified AI system is trusted more than an uncertified AI system. Likewise, users rely more on a certified system than an uncertified system.

Building on previous works that has found negative effects of certifications, we suspect that similar processes might play out for AI certifications. To that end, we suggest that certifications might raise users' expectations beyond what a certification can guarantee. With system performance as a key predictor [49] of users' trust, this should be particularly evident when a system performs less well. To examine this, we predicted that:

H3: An AI system with high reliability is (a) trusted and (b) relied on more than an AI system with low reliability.

H4: For an AI system with high reliability, trust in an AI system does not differ between a certified AI system and an uncertified AI system, indicating the proposed ceiling effect by Wischnewski et al. [102]. For an AI system with a lower reliability, however, a certified AI system is (a) trusted and (b) relied on less than an uncertified AI system.

Linking this to expectation violations, we further suggest that:

H5: For an AI system with a lower reliability, the negative effect of the certification on trust and reliance in an AI system is mediated by negative expectation violations. Thereby, a certified AI system with a lower reliability elicits more negative expectation violations than an uncertified AI system with a lower reliability.

3 Method

The present study received ethical approval from the ethics committee of the University of [anonymized for review] under ID 2411SPSA9515. All hypotheses and analyses were preregistered and can be found here. All data are stored here.

3.1 Experimental Design

To test our hypotheses, we developed a 2 (certification vs. no certification) x 2 (reliability: high vs low) between-subjects design to which N = 644 participants were randomly assigned.

Certification manipulation: In all conditions, we first introduced participants to the process of AI certifications, addressing the need for certification and the criteria required to obtain it (see Figure 1). Because the design of the certification mark can impact users' judgments (see e.g., [39]), similar to [102], we told participants that the actual certification mark could not be displayed and the picture shown was a placeholder.

In the experimental condition, participants were told that they were about to interact with a certified system. To reinforce this treatment, whenever participants received assistance from the system, we reminded participants that the AI was certified, and added the certification mark. In contrast, participants in the control condition were told that the system they would be interacting with was not certified. After each interacting with the system, this was repeated.

Reliability manipulation: We manipulated the reliability of the AI system by varying the number of incorrect recommendations that participants received. Hence, participants in the low-reliability condition received more incorrect recommendations from the system than those in the high-reliability condition. Overall, the errors consisted of both overestimations and underestimations, each deviating from the correct percentage by 11 to 15 percentage points. A detailed overview of the errors for both reliability levels across all ten estimation questions is provided in the Supplementary material A.

3.2 Task

To answer our hypotheses, we asked participants to complete a two-step estimation task with the help of the alleged AI system, QuantifierAI. Participants were told to imagine that they were working in a laboratory specializing in the diagnosis of bacterial infections. Their task was to identify bacterial contamination in different samples (see Figure **??**. To estimate the area of bacterial infection, they received the support of QuantifierAI. Depending on the experimental condition, QuantifierAI was either accompanied by a certification mark or not.

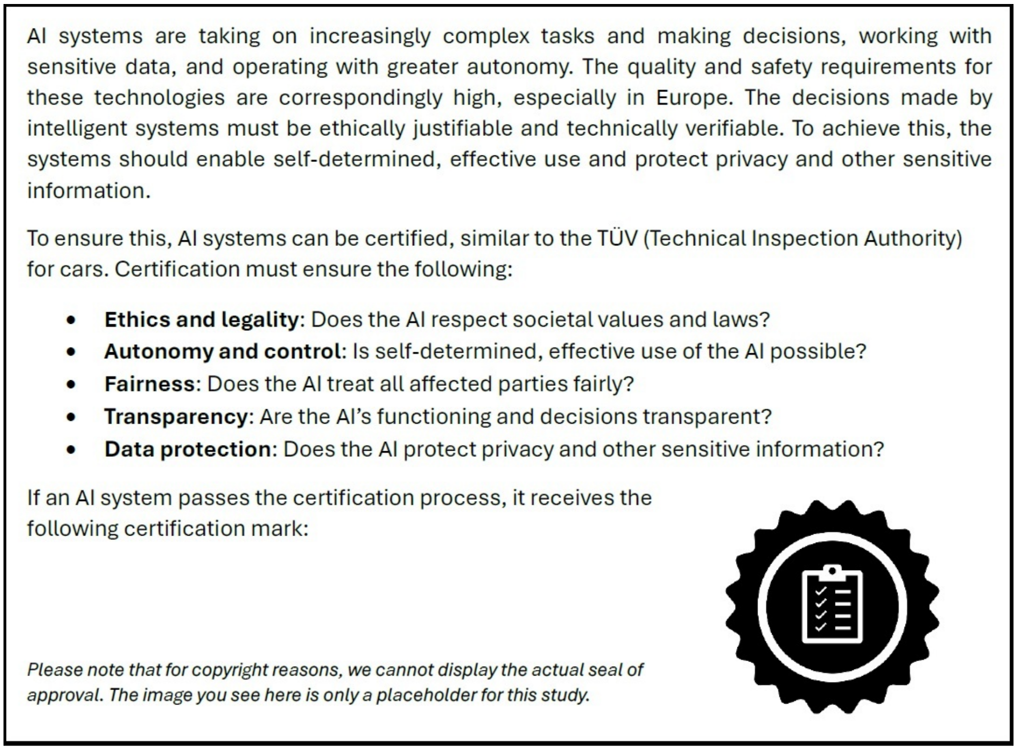
All participants completed a total of ten estimation questions. Each estimation question followed a two-step approach. Participants first provided their own estimation (in percentage), and received a recommendation from the AI system, accompanied by the note: "You now have the option to adjust your answer or leave it as it is." Participants then selected their final answer and were taken to the feedback page. There, they received feedback regarding the accuracy of their final answer, including the correct answer and whether QuantifierAI's recommendation was correct or incorrect. Answers were correct when they were within a tolerance range of three percent above or below the correct answer.

As Wischnewski et al. [103] noted, selecting an appropriate task difficulty is important when examining trust and reliance. If tasks are too easy, participants have little incentive to depend on a system. In contrast, if tasks are too difficult, users may choose to always follow the AI advice, not because they trust but because they have no other choice but to use the system, as they do not possess the skills themselves. To that end, visual estimation tasks are not only well established in HCI and perceptual research, but have also been found to reliably produce measurable accuracy rates of roughly 30% [23, 41, 86].

Moreover, this two-step estimation task ensured that we could capture both behavioral reliance and subjective trust by simultaneously reducing possible effects of prior domain knowledge, thereby reducing confounds related to expertise. Additionally, the laboratory-diagnosis scenario provided situational realism without introducing ethical risks. The task and all materials were directly sourced and adopted from Fahnenstich et al. [34].

3.3 Procedure

After providing informed consent, participants completed a short demographics questionnaire and were introduced to the study context, including a brief explanation of the certification process relevant to the AI system (see Figure 1). They then received instructions about the estimation task and the role of the AI assistant, QuantifierAI. Before interacting with the system, participants responded to measures of trust in as well as expectations and vigilance towards QuantifierAI. The interaction phase consisted of ten estimation trials in which participants first made their own estimate, viewed the AI's recommendation, and submitted their final answer. After each trial, participants received feedback on their final answer and whether the AI recommendation was correct. Following the interaction phase, participants again completed the expectation, trust, and



AI systems are taking on increasingly complex tasks and making decisions, working with sensitive data, and operating with greater autonomy. The quality and safety requirements for these technologies are correspondingly high, especially in Europe. The decisions made by intelligent systems must be ethically justifiable and technically verifiable. To achieve this, the systems should enable self-determined, effective use and protect privacy and other sensitive information.

To ensure this, AI systems can be certified, similar to the TÜV (Technical Inspection Authority) for cars. Certification must ensure the following:

- **Ethics and legality**: Does the AI respect societal values and laws?
- **Autonomy and control**: Is self-determined, effective use of the AI possible?
- **Fairness**: Does the AI treat all affected parties fairly?
- **Transparency**: Are the AI's functioning and decisions transparent?
- **Data protection**: Does the AI protect privacy and other sensitive information?

If an AI system passes the certification process, it receives the following certification mark:

Please note that for copyright reasons, we cannot display the actual seal of approval. The image you see here is only a placeholder for this study.

Figure 1: Information presented to all participants addressing (a) the need for AI certifications, and (b) the criteria required to obtain a certification.

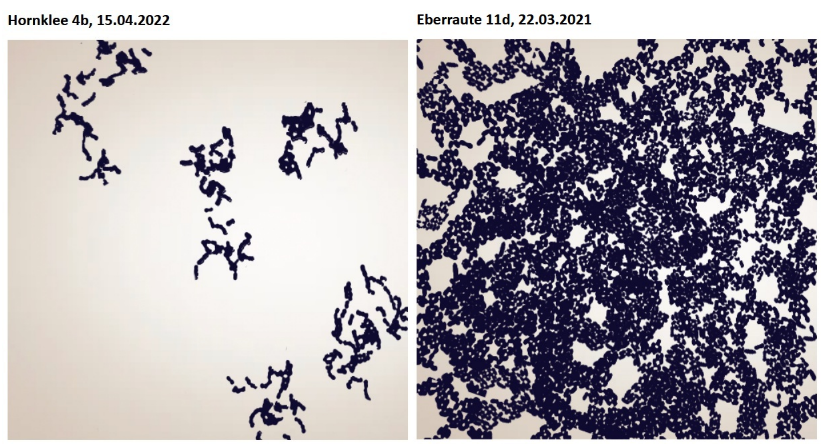


Figure 2: Example images, displaying either low infection (left) of 7%, and high infection (right) of 72%.

vigilance measures. Finally, all participants were debriefed about the study purpose.

3.4 Sample

To determine the sample size, we ran an a priori power analysis (effect size f = .15, significance level .05, & test power of .95) using G*Power that revealed a sample size of 580 participants. Based on this, we collected data from N = 651 participants via the online panel provider Prolific. All participants were compensated for their participation. Participants were eligible if they were above the age of 18, had to have a good understanding of [anonymized for review], the language in which the experiment was carried out, and an acceptance rate of at least 98% on Prolific. After data screening, we excluded seven participants due to failed manipulation and attention checks (see more on manipulation and attention checks in

Table 1: Demographic characteristics of participants.

Variable	Value
Gender: Female / Male / Diverse	331 / 306 / 7
Age: Range	18–74
Age: Mean (SD)	35.39 (11.45)
Education: University degree	376
Education: High school / university entrance	143
Education: Secondary / vocational diploma	52
Education: Specialized high school / technical college	48
Education: Lower secondary or equivalent	10
Education: No diploma	4
Education: Other	11
Familiarity with AI certification: Yes / No	112 / 532

section 3.6). We provide an overview of participants' demographic data in Table 1.

3.5 Measures

3.5.1 Trust in the AI system. Trust in the AI system was assessed using the general trust dimension of the Trust in AI Scale (TAIS) by [101] that consists of the five items "I believe in QuantifierAI," "I trust QuantifierAI," "I distrust QuantifierAI," "QuantifierAI is designed to be trustworthy," and "I can depend on QuantifierAI." The respective statements were rated on a five-point Likert scale (1 = strongly disagree to 5 = strongly agree). Trust was measured both after reading the introductory text before the interaction phase (pre-interaction measurement) and after the interaction phase (post-interaction measurement). The scale was very reliable (Cronbach's alpha pre = .868, Cronbach's alpha post = .916).

3.5.2 Vigilance. In addition to trust, we also measured users' vigilance toward the system to gain more methodological depth. Vigilance toward the AI system was assessed using the vigilance dimension of the Trust in AI Scale (TAIS) by [101], consisting of the five items "I am vigilant about QuantifierAI," "I am alert of QuantifierAI," "I am on the lookout when using QuantifierAI," "I am on my toes when using QuantifierAI," and "I am careful when using QuantifierAI." Responses were assigned to a five-point Likert scale (1 = strongly disagree to 5 = strongly agree). Vigilance was measured both after reading the introductory text before the interaction phase (pre-interaction measurement) and after the interaction phase (post-interaction measurement). The scale was very reliable (Cronbach's alpha pre = .874, Cronbach's alpha post = .925).

3.5.3 Reliance. Reliance on the AI system was calculated using the Weight of Advice (WoA), a widely used measure for instrumentalizing the use of recommendations [59, 105]. For each interaction, the impact of the AI recommendation on participants was calculated as follows:

$$\text{WoA} = \frac{\text{finale answer} - \text{initial answer}}{\text{AI recommendation} - \text{initial answer}}$$

Thus, the WoA reflects the percentage weight that participants assign to the AI's recommendation [105]. This can provide insight into the extent to which the AI's recommendation influenced the person's final decision and thus the strength of any changes in their decision. The WoA is limited to a range from -1 to 1, allowing a maximum adjustment of 100% to the AI's recommendation [59]. Accordingly, it takes the value 0 if participants stick with their initial answer and 1 if they accept the AI's recommendation as their final answer [105]. A negative value implies that participants moved further away from the AI's recommendation, while a positive value implies that they have moved closer to it [59]. For the analyses, we calculated the mean WoA over all ten estimation decisions.

3.5.4 Expectation violations. We used an indirect and direct approach to assess participants' expectation violations, to enable more robust results. For the indirect measurement, we followed Crolic et al. [26], Zhang et al. [109], who used a pre-interaction expectation measurement and a post-interaction evaluation. The expectation violation score was then calculated by subtracting the post-interaction evaluation from the pre-interaction expectations [26]. To that end, we used the ability dimension from Wischnewski et al. [101] to assess participants' expectations with the items "QuantifierAI fulfills its task very well," "QuantifierAI is capable," "QuantifierAI is proficient," "QuantifierAI functions properly," and "QuantifierAI responds accurately", rated on a five-point Likert scale (1 = strongly disagree to 5 = strongly agree). In the pre-interaction measurement (expectations), participants were instructed to refer to their expectations of QuantifierAI. In the post-interaction measurement (evaluation), they were asked to refer to their experiences with QuantifierAI and how they perceived QuantifierAI. Possible values ranged from -4 (strong positive expectation violations) to +4 (strong negative expectation violations), with a value of zero corresponding to no expectation violations. The scales were reliable (Cronbach's alpha pre = .888, Cronbach's alpha post = .951).

In addition, we developed five items to directly assess participants' expectations after the interaction: "QuantifierAI disappointed me," "I had expected better performance from QuantifierAI," "QuantifierAI was less helpful than I had assumed," "I had thought QuantifierAI would work better," and "I had expected better results from QuantifierAI." Participants rated the statements on a five-point Likert scale (1 = strongly disagree to 5 = strongly agree). The scale was very reliable (Cronbach's alpha = .971).

3.6 Manipulation and attention checks

To ensure high data quality, we implemented two attention checks by asking participants twice to select a specific answer within the questionnaire. As a result, n = 7 participants were excluded from the analysis. Additionally, to gain a better understanding of users' awareness of the presence of a certification, by the end of the survey, we asked participants to complete the following sentence: „QuantifierAI was…“ by choosing either "certified", "not certified", or "I don't know". We used a Chi-square test to analyze if the number of wrong answers was significant. The results $\chi^2(2, 644) = 538.70$, $p < .001$ indicated that participants mostly remember whether or not a certification was displayed (correct answers: not certified = 298, certified = 306).

4 Results

The statistical analyses reported below were performed using Jamovi statistical software version 2.3.21. A significance level of 5% was used to test significance. The descriptive results by experimental conditions and divided into pre and post interaction with the system can be found in Table 2 and 3.

Table 2: Descriptive Statistics of the pre-interaction levels of trust, expectations, and vigilance by experimental condition.

Variable	not certified		certified		Total	
	M	*SD*	*M*	*SD*	*M*	*SD*
trust	3.19	0.65	3.53	0.67	3.36	0.69
expectation	3.47	0.60	3.78	0.64	3.63	0.64
vigilance	3.87	0.69	3.67	0.69	3.77	0.70

4.1 RQ1: How do AI certifications shape users' pre and post-interaction perceptions of the system?

In H1, we predicted that, before an actual interaction with the system, the certified AI system is trusted more than the uncertified system. To test this, we ran two separate analyses of variance (ANOVA) with the pre-interaction levels of trust in and vigilance towards the AI system as the dependent variable and the grouping variable certification as the independent variable. For trust in the AI system, our results indicated that the certification had a significant, medium-sized effect on trust ratings ($F(1, 642) = 42.96$, $p < .001$, $\eta^2 = .06$). Before an interaction, the AI system was significantly more trusted when certified ($M = 3.53$, $SD = 0.67$) than when not certified ($M = 3.19$, $SD = 0.65$), supporting H1.

Similar to the trust ratings, the results of the second ANOVA showed that, before interacting with the AI system, the certification had a significant and small effect on vigilance toward the AI system, with $F(1, 642) = 13.38$, $p < .001$, $\eta^2 = .02$. Before the interaction with the system, participants were more vigilant toward the uncertified AI system ($M = 3.87$, $SD = 0.69$) than the certified system ($M = 3.67$, $SD = 0.69$).

In hypothesis 2, we predicted that the significant effect of the certification on trust and vigilance remained after the interaction. Additionally, we suspected that the certification should also affect how users rely on a system, with greater reliance on a certified than an uncertified system. To test this hypothesis, we ran three separate ANOVAs, each with the experimental manipulations certification and reliability as independent grouping variables and (1) trust, (2) vigilance, and (3) reliance as dependent variables.

The results of the ANOVA indicated that, after an interaction, the effect of the certification on trust vanished ($F(1, 640) = 0.22$, $p = .641$, $\eta_p^2 < .001$). The certified ($M = 3.04$, $SD = 1.05$) and uncertified system ($M = 3.01$, $SD = 0.94$) elicited similar levels of trust. Likewise, the certification did not affect vigilance levels significantly anymore, with $F(1, 640) = 1.16$, $p = .282$, $\eta_p^2 = .002$. The certified ($M = 3.84$, $SD = 0.87$) and uncertified system induced similar levels of vigilance ($M = 3.91$, $SD = 0.79$). Inspecting users' reliance on the AI system, however, we found a significant, small effect of the certification, with $F(1, 640) = 5.96$, $p = .015$, $\eta_p^2 = .009$. Participants interacting with the certified system relied significantly more on the system ($M = 0.60$, $SD = 0.23$) than participants interacting with the uncertified system ($M = 0.55$, $SD = 0.23$). Considering the results of all three ANOVAs, we have to reject H2 for trust and vigilance: After interacting with the AI system, the certification had no effect on trust and vigilance levels. However, we found support for H2 for reliance: After interacting with the AI system, the certification affected reliance. Yet, the effect size was very small.

4.2 RQ2: How do certifications in combination with the system's performance shape users' trust and reliance?

In a second step, we inspected the effect of the second experimental manipulation, reliability. We hypthesized that users would trust a reliable system more and would be less vigilant toward a reliable system than an unreliable system (H3). We found a large effect of reliability on all three dependent variables. Participants trusted the reliable system significantly more ($M = 3.70$, $SD = 0.66$) than the unreliable system ($M = 2.34$, $SD = 0.78$) ($F(1, 640) = 578.01$, $p < .001$, $\eta_p^2 = .47$). Likewise, the reliability had a significant and moderate effect on vigilance toward the AI system after interacting with it ($F(1, 640) = 101.22$, $p < .001$, $\eta_p^2 = .137$). Participants became significantly more vigilant toward the system when its reliability was low ($M = 4.18$, $SD = 0.74$) than when it was high ($M = 3.57$, $SD = 0.80$). Lastly, reliability significantly affected participants' reliance on the system, with $F(1, 640) = 4.68$, $p = .032$, $\eta_p^2 = .01$. We conclude that, after interacting with a system, the performance in terms of reliability affected users' trust, vigilance and reliance more than a certification.

We also suspected that the certification might raise users' expectations to a degree that the certification cannot guarantee. In turn, this would result in an expectation violation when the system is less reliable. To test this, we inspected the interaction effect of the two experimental manipulations, certification and reliability on trust, vigilance, and reliance. Concretely, in H4, we predicted that an unreliable but certified system would be trusted less, would induce higher vigilance and less reliance than an unreliable, uncertified system.

Like hypothesized, for trust, the interaction term indicated a small, significant effect ($F(1, 640) = 6.01$, $p = .014$, $\eta_p^2 = .009$). To further scrutinize the direction of the effect, we conducted post-hoc comparisons with Bonferroni-corrected alpha levels. However, unlike hypothesized, an unreliable system with a certification was not trusted less than an unreliable system without certification ($t(640) = 1.40$, $p = .500$, d = 0.16).

To better understand the significant interaction term, we additionally explored the other comparisons. Supporting the main effect, we found a significant difference between unreliable and reliable systems ($t(640) = 15.27$, $p < .001$, d = 1.70) in terms of trust levels. Interestingly, this effect was more pronounced for certified systems ($t(640) = 18.73$, $p < .001$, d = 2.09), letting us concluded that the difference in trust in the AI system between the AI system with high reliability and the AI system with low reliability appeared to be greater when a system is certified than when it is not (see Figure 3 for a visualization).

For vigilance, we found no such interaction effect ($F(1, 640) = 0.39$, $p = .532$, η2p = .001). Likewise, for reliance, the interaction was not significant ($F(1, 640) = 1.57$, $p = .211$, η2p = .002).

Table 3: Descriptive statistics of the post-interaction levels of trust, evaluation, expectation violations (EV) (direct and indirect measurement), and vigilance by experimental condition.

	Not Certified				Certified				Total	
	High Reliability		Low Reliability		High Reliability		Low Reliability			
Variable	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
trust	3.62	0.66	2.39	0.76	3.78	0.66	2.28	0.79	3.02	0.99
evaluation	3.97	0.62	2.53	0.75	4.12	0.55	2.35	0.80	3.25	1.06
EV (indirect)	-0.51	0.66	0.96	0.93	-0.33	0.65	1.42	0.99	0.38	1.16
EV (direct)	1.73	0.78	3.84	1.00	1.79	0.89	4.08	0.92	2.86	1.42
vigilance	3.62	0.77	4.20	0.70	3.52	0.84	4.17	0.78	3.87	0.83

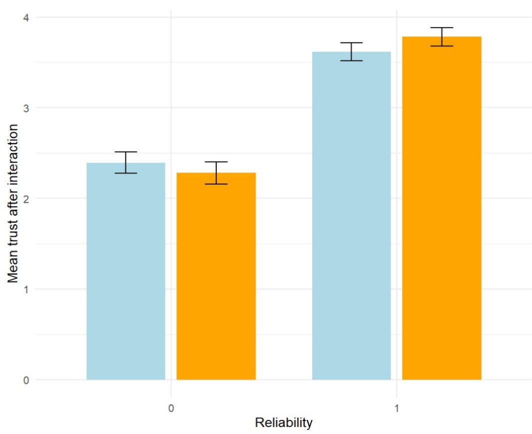


Figure 3: Mean trust ratings by all four experimental conditions.

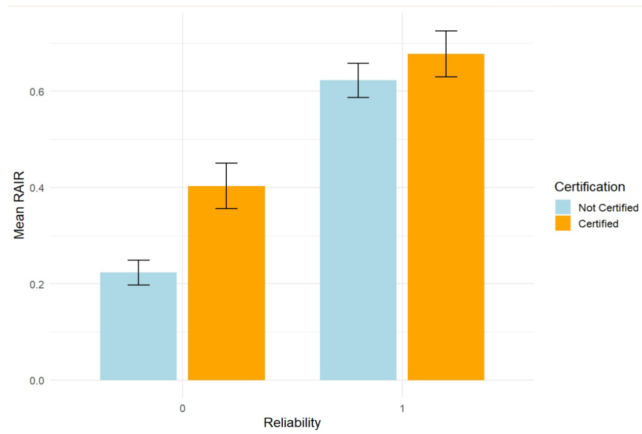


Figure 4: Mean RAIR by all four experimental conditions.

We extended these pre-registered analyses by additionally exploring possible effects on the appropriateness of users' reliance. While we did not anticipate any effects a priori, during the analysis, it became apparent that reliance alone was not sufficient. Hence, following Schemmer et al. [82], we calculated participants' Relative AI Reliance (RAIR) and Relative self-reliance (RSR) to gauge participants' appropriate reliance on AI advice. RAIR represents the proportion of trials in which participants initially provided an incorrect response, received correct AI advice, and subsequently changed their response to the correct answer, relative to all trials in which the participant was initially incorrect and the AI advice was correct, regardless of the final decision. RSR represents the proportion of cases in which participants were initially correct, received incorrect AI advice, and still kept their correct answer, divided all instances in which the participant initially was correct, and the AI was incorrect, reflecting appropriate refusal to follow faulty AI suggestions. RAIR and RSR values range from 0 to 1, with higher values indicating more appropriate reliance on the AI.

Following the procedure for trust, vigilance, and reliance, we conducted a two-way ANOVA to examine the effects of AI reliability and certification on mean RAIR per participant over all trials. We found a large effect of AI reliability on RAIR. Participants correctly relied on the reliable system significantly more (M = 0.651, SD = 0.272) than on the unreliable system (M = 0.313, SD = 0.238) ($F(1, 639) = 270.18$, $p < .001$, $\eta_p^2 = .30$). There was also a significant effect of certification, with $F(1, 639) = 32.70$, $p < .001$, $\eta_p^2 = .05$. Participants interacting with the certified system relied more often accurately on the AI (M = 0.541, SD = 0.308) than when interacting with the uncertified system (M = 0.423, SD = 0.241). The interaction between reliability and certification was also significant, $F(1, 639) = 9.30$, $p = .002$, $\eta_p^2 = .015$. To better understand this finding, we conducted simple-effects analyses. We found that, when the system was unreliable, the certification significantly supported users to only follow correct AI advice, $t(639) = -6.20$, $p < .001$. When the system was reliable, the effect of certification was smaller and only approached significance, $t(639) = -1.90$, $p = .058$ (see Figure 4 for a visualization). These findings suggest that the certification primarily improves reliance on correct AI advice (and not incorrect) when the AI system is less reliable, with a smaller effect when the system is already reliable.

RSR was only computable for participants who encountered at least one trial where they were initially correct, and the AI was wrong. As we assumed relatively high task difficulty, we first inspected how often this was the case (NA per condition). Here it stood out that for the lower reliability conditions, when participants were correct and the AI was wrong, they stayed correct in about 30% of the time (not certified: n = 22, M = .273; certified: n = 101, M= .316), reflecting previously reported accuracy rates of visual estimation tasks [23, 41, 86]. Notably, when the system was not certified and reliability was high, participants showed worse RSR (n = 17, M = .176). Yet, when the system was certified and reliable RSR was highest (n = 110, M = .889), indicating that when the system was certified, participants were exceptionally good at rejecting incorrect AI advice when it occurred. Because of the varying n of RSR throughout the conditions, we did not perform any additional ANOVA, but report descriptive statistics and interpret patterns cautiously.

4.3 Can negative expectation violations account for the observed interaction between certification and reliability on trust

While we only found a significant interaction effect for trust and not for vigilance and reliance, we wanted to know what could explain this effect. Hence, in hypothesis H5, we suspected that, when interacting with a certified AI system that is less reliable, users would experience greater negative expectation violations than with a system that is not certified.

To test for this, we conducted two separate ANOVAs with the two experimental conditions as grouping variables, and the two versions of the expectation violation assessment (direct and indirect assessment) as dependent variables, respectively. We did not consider the main effects in our hypotheses but, for the sake of completeness, provide a full report here nevertheless.

For the indirect measure of the expectation violation, we found a significant main effect of the certification ($F(1, 640) = 24.64$, $p < .001$, $\eta_p^2 = .04$) with the certified system inducing more expectation violation than the uncertified system (certified: $M = 0.54$, $SD = 1.21$; uncertified: $M = 0.22$, $SD = 1.09$). Likewise, the main effect of reliability on the indirect expectation violation measure was also significant (F(1, 640) = 623.97, p < .001, $\eta_p^2 = .49$). The high reliability was associated with a positive expectation violation ($M = -0.42$, $SD = 0.66$), while the low reliability was associated with negative expectation violations ($M = 1.19$, $SD = 0.99$). Both main effects were not hypothesized, but became apparent during the analysis.

Inspecting the interaction effect of both experimental variations, certification and reliability, we found a small, significant interaction effect for the indirect assessment of expectations, with $F(1, 640) = 4.67$, $p = .031$, $\eta_p^2 = .007$. Post-hoc test revealed that for the reliable system, expectations violations did not differ between the certified and uncertified system ($t(640) = -1.99$, $p = .193$, d = -0.22). In turn, for the less reliable system, we found a significant difference in expectations violations between the certified and the uncertified system ($t(640) = -5.02$, $p < .001$, d = -0.56), with the certified system inducing more expectations violations, supporting our hypothesis.

The direct measurement of the expectations violation partly supported these results: We found a small but significant main effect of the certification ($F(1, 640) = 4.65$, $p = .031$, $\eta_p^2 = .007$). Similar to the indirect measure of the expectation violation, when keeping reliability constant, the certified system ($M = 2.93$, $SD = 1.46$) induced more expectation violation than the uncertified system ($M = 2.78$, $SD = 1.38$). The reliability of the AI system had a significant and large effect with $F(1, 640) = 960.52$, $p < .001$, $\eta_p^2 = .60$, whereby the AI system with low reliability ($M = 3.96$, $SD = 0.97$) triggered significantly more negative expectation violations than the AI system with high reliability ($M = 1.76$, $SD = 0.83$).

Unlike the indirect measure, the interaction of the certification and the reliability condition was not significant ($F(1, 640) = 1.74$, $p = .187$, $\eta_p^2 = .003$).

While we found no direct effect of the certification on trust after the interaction with the system, we conducted a mediation analysis to test for a possible mediation, following recommendations by O'Rourke and MacKinnon [73], Zhao et al. [110]. To test for this, we conducted a mediation analysis with a 5000 bootstrap sample, with the certification as the independent variable, trust as the dependent variable, and the indirect measure of the expectation violation as the mediator.

Supporting the results of the ANOVA, we found that the certification had no significant effect on trust in the AI system after interacting with it (c = -0.113, β = -0.073, z = -1.303, $p = .193$). This did not change after expectation violations were included as a mediator (c' = 0.082, β = .053, z = 1.061, $p = .289$). However, a significant indirect effect of the presence of the certification on trust in the AI system was found via expectation violations (ab = -0.195, β = -0.126, z = -3.810, $p < .001$). Given the low reliability of the AI system, the certification led to more negative expectation violations than no certification, which subsequently reduced trust in the AI system. Figure 5 shows the results of the mediation analysis. Thus, the mediation was present, and H5 was supported for the indirect measurement of expectation violations (see Figure 5).

Next, we conducted a second mediation analysis; this time with the direct measure of the expectation violation measure as mediator. The results of the second mediation analysis supported the findings of the first. There was no significant effect of the certification on trust in the AI system (c = -0.113, β = -0.073, z = -1.314, $p = .189$). Similarly, we did not find an effect of the certification when including the expectation violations as a mediator (c' = 0.006, β = .004, z = 0.087, $p = .930$). However, we found an indirect effect of the certification on trust in the AI system (ab = -0.118, β = -0.077, z = -2.282, $p = .023$). Given the low reliability of the AI system, the certification led to more negative expectation violations than the absence of the certification, which subsequently reduced trust in the AI system (see Figure **??** for a visualization).

5 Discussion

As more and more AI regulatory frameworks introduce AI certifications, the aim of this paper was to gain a deeper understanding of how such AI certifications possibly influence users' trust in and reliance on AI systems. Against this backdrop, we investigated the effect of a certification at different points in time on trust in and reliance on an AI system. Additionally, we manipulated the reliability of the AI system to understand how the system's behavior possibly affects any effect of the certification. We were particularly interested whether a certified but less reliable system would elicit negative expectation violations that would subsequently affect trust and reliance.

5.1 Results of the hypotheses testing

Before interacting with a system, certifications raise trust and lower vigilance towards the system. As hypothesized, the certification had a significant effect on trust and vigilance prior to the interaction with the system. Our participants trusted the certified system more and were less vigilant towards the certified system as compared to the uncertified system.

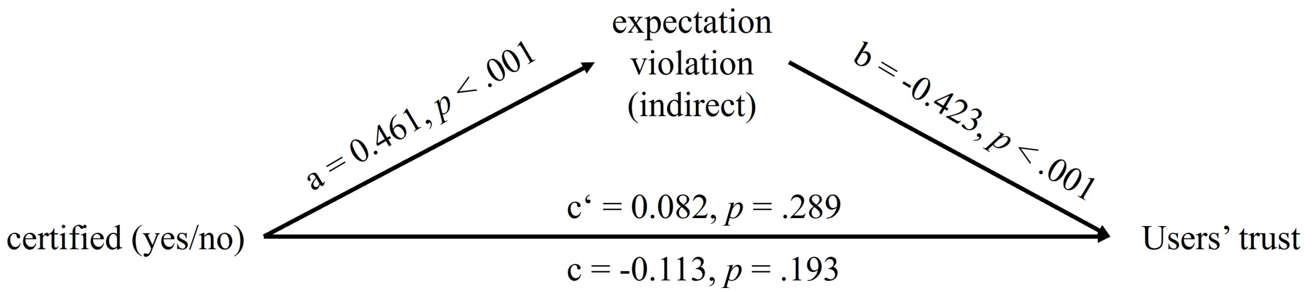


Figure 5: Results of the mediation analysis, using the indirect measure of expectation violations as a mediator.

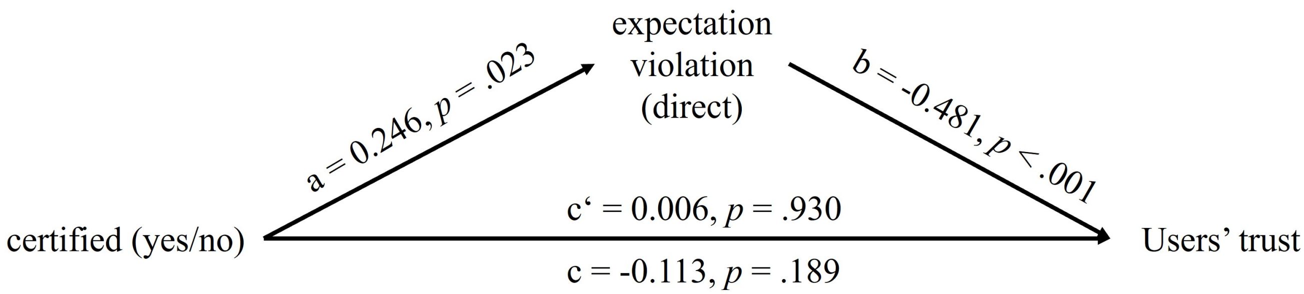


Figure 6: Results of the mediation analysis, using the direct measure of expectation violations as a mediator.

This finding aligns well with prior empirical research suggesting that signals of legitimacy, such as certifications, can act as signals that boost trust in unfamiliar technologies [15, 52, 93]. Likewise, our results align with previous work on AI certifications that could show that AI certifications increased users' trust, acceptance, and their willingness to use the AI [27, 81].

The certification, by indicating that the system meets certain predefined standards, may have acted as a surrogate for the user's lack of information, encouraging them to trust the system at face value. In the context of AI, this effect may be particularly pronounced as users often rely on simple signals to judge complex technologies with which they have limited experience [80].

Ultimately, to better understand the effect of the certification, future studies are needed to further investigate the underlying psychological mechanisms that explain this effect. For example, if the AI certification serves as a heuristic cue, signaling that the system has met certain standards or undergone evaluation, this should reduce users' cognitive load [83]. In turn, various constructs related to cognitive load should be affected, such as user satisfaction [46] and confidence [111]. In addition, certification marks may trigger other biases, such as authority or halo effects in reference to the certifying instance [102], or interact with participants' prior attitudes toward AI [4], which could influence reliance independently of cognitive load or task difficulty.

After interacting with a system, certifications had no effect on trust in and vigilance towards the system but on reliance and appropriate reliance. The effect of the AI certification on trust and vigilance did not persist after users interacted with the system. Instead, the system's performance in terms of reliability emerged as a decisive factor in shaping trust and vigilance (H3). This shift from an initial reliance on certifications to reliance on performance aligns with theories of trust that emphasize the dynamic and evolving nature of trust in technology. For example, Kaplan et al. [49] found in their meta-analysis that AI-related factors had the strongest influence on trust, with reliability emerging as the strongest predictor. Likewise, Wischnewski et al. [101] found that, in their model of trust in AI, general trust levels could be largely explained by users' perceptions of the system's ability.

Moreover, the evolvement of trust in our findings corresponds to the assumptions of the framework developed by Hoff and Bashir [44], which distinguishes between initially learned trust (before interaction) and dynamically learned trust (during interaction). While initially learned trust is influenced by existing knowledge, such as provided by a certification, dynamically learned trust is largely determined by perceived system performance, which includes reliability, among other things.

Overall, this suggests that, while certifications can initially boost users' trust, trust is ultimately more influenced by users' experience with the system. From a normative perspective, this can be viewed as positive, as it indicates a differentiated and well-founded development of trust that is presumably less guided by potentially superficial external characteristics and more dependent on an informed assessment of actual interaction and the experiences gained in the process [103].

Nevertheless, we found that, while showing no effect for trust and vigilance, the certification affected participants' reliance behavior. Through our pre-registered hypothesis testing, we found that participants relied more on the certified than the uncertified system. Adding to this, we also explored participants' appropriate reliance behavior. Our results show that certifications affected both reliance and appropriate reliance. This highlights two things: First, although trust and reliance are often associated with each other [29], with greater trust leading to greater reliance, they are not identical, and can diverge when contextual cues selectively shape behavior without altering underlying beliefs. Differentiating between trust and trusting behavior, such as reliance is crucial (see for an in-depth discussion Patton and Wickens [74]).

Second, as the certification affected reliance but not trust, we speculate that other mechanisms were at play. We surmise that the certification may have functioned as a sort of authority heuristic: a mental shortcut where people rely on the opinions of authority figures or experts to make decisions [88]. Prior work supports the idea that such cues shift behavior independently of internal evaluations. For example, Sundar et al. [89] found that, next to peer recommendations, authority cues in the form of seals shaped product attitudes and purchase intentions, especially when peer cues were inconsistent. Although their work examined consumer judgments, the same mechanism plausibly applies to AI systems, where institutional markers such as certifications signal compliance with regulatory standards.

Our results on appropriate reliance show that the certification enabled users to rely on the system more appropriately, particularly when the system was unreliable. While relative AI reliance (RAIR) was higher for reliable systems overall, the certification boosted RAIR most strongly in the context of lower reliability. We speculate that the large effect of the certification for the less reliable system may be caused by the difficulty of the task: participants struggled to perform well on their own and thus benefited from following the AI's recommendations. However, when the AI was less reliable, participants under-relied on it, possibly reflecting participants' aversion to AI errors [17, 48]. The certification mitigated this under-reliance, guiding participants to make more appropriate use of the system's advice and improving decision accuracy.

For relative self-reliance (RSR), the pattern was complementary: Participants struggled to maintain correct responses when the AI provided incorrect advice, except in the high-reliability, certified condition. We understand this effect as the certification increasing users' sensitivity to rare AI errors, promoting appropriate self-reliance. However, the small and highly uneven group sizes for RSR precluded formal inferential testing, and future studies should examine this more closely.

Certifications exacerbate the effect of reliability. In H4, we hypothesized that the certification might induce negative expectation violations by raising expectations to an extent that the certification does not account for. We expected this to be particularly pronounced when the system performs less well.

Although the interaction effect did not turn out as assumed in the hypothesis, a more detailed exploratory analysis showed that the effect of reliability appears to be reinforced by a certification. The difference in trust after interacting with the AI system between the AI system with high reliability and the AI system with low reliability was greater when the AI system was certified.

As described in our hypothesis, for systems with low reliability, we suggest that the certification was detrimental to trust because users might overinterpret what a certification can guarantee. If a system does not perform well but it is certified, users trust the system less than without the certification. Our findings regarding the expectation violations support this idea (see more in the next section). In turn, for the highly reliable system, the certification boosted the effect of the high reliability. While not testing for this, we suspect that the certification induced some sort of a halo effect [71], where users' positive impression of the certification (see pre-interaction increases in trust) affected their perceptions of the reliability.

Certifications elevate user expectations about the system performance. Corroborating the results found for the effects of the certification for different reliability levels, users' expectations changed when interacting with a certified system, depending on the system's reliability. More concretely, our findings suggest that certifications might have elevated user expectations, making them more sensitive to deviations from these expectations. While we did not find that, when the system performed reliably, the certification affected expectations, we found that, when the system performed unreliably, the certification induced more negative expectations violations.

This result was later supported by the mediation analysis (H5). There, we found an indirect effect of negative expectations violations on users' trust, without a total effect of the certification on trust levels. Following Zhao et al. [110] on the interpretation of indirect-only mediations, the hypothesized mediation of negative expectation violations was confirmed. However, because the total effect of the certification on trust was non-significant, it might have been possible that another, additional opposite mediator existed. The opposite signs of both mediators in combination would cancel out the total effect. Similar results have been observed, for example, by Waddell [99] who found that the effect of machine authorship on credibility ratings was masked by two competing mediating variables: perceived bias and source anthropomorphism.

We suggest that another indicator for an opposite mediator is presented by the positive effect of the certification on reliance and appropriate reliance. In other words, while the certification benefited (appropriate) reliance, it had a negative effect on trust. Hence, a possible mediator would have to affect trust and reliance positively. This could be, for example, perceived uncertainty (see e.g., [53]) that is reduced by the certification.

Overall, the results of the mediation highlight that it is important to go beyond the mere presence of direct effects, but to consider possible underlying psychological mechanisms.

5.2 Practical implications

Our results indicate that certifications represent an effective way to increase initial trust in AI systems and reduce initial vigilance toward them. This effect would be particularly beneficial in procurement processes, where decision-makers often need assurance regarding the quality, safety, and ethical standards of AI systems before deployment. By providing a third-party verification of AI performance and compliance with ethical standards, certifications help mitigate uncertainty and reassure stakeholders, ultimately streamlining decision-making and fostering adoption. This could be especially crucial in industries like healthcare or finance, where AI systems' impact is profound, and trust is a central concern.

However, our results also suggest that AI certifications are less likely to influence trust and vigilance, once a system is in use. Our findings indicate that this was due to the strong effect of the system's performance in terms of reliability, showing that users may develop a more nuanced understanding of the system's limitations and capabilities, and thus become more critical, regardless of certification status.

Critically, in line with previous works [55], our findings show that users may have over-generalized certifications, extending beyond the scope of what the certification actually covers. This finding highlights the importance of setting realistic expectations around the role of certification. While certifications play a crucial role in building initial trust, they should be viewed as just one part of a broader strategy for ensuring system transparency, accountability, and continuous improvement. It also underscores the need for clearer communication regarding the scope and limitations of certifications, so users do not over-rely on them or assume they offer an ongoing guarantee of system performance. Further, it suggests that organizations issuing certifications should consider providing more detailed information about the specific aspects of AI systems being certified and the conditions under which the certification is valid, to help users better understand its scope and limitations.

This is easier said than done. For the context of AI energy labels, Fischer et al. [36] found, for example, that, while users valued simplicity of a label to improve accessibility and comprehension, they also wanted more detailed information, indicating a simplicity-complexity trade-off. Yet, studies evaluating security designs favor information-dense labels, which were found to create more secure purchases [69, 79]. Applying this to the context of AI certifications, setting the right expectations to avoid over-generalizations necessitates more detailed information about what the certification guarantees, increasing complexity, but allowing users to create more accurate perceptions. This might change, however, once a certification is more established [16].

5.3 Limitation

While this study contributes insights into the role of certifications in shaping trust and reliance towards AI systems, several limitations should be considered.

As mentioned above, one important limitation is the absence of a measure for appropriate reliance on AI certifications. Although our findings suggest that certifications can boost initial trust in AI systems, they do not capture the degree to which users should actually rely on these certifications when making decisions. Likewise, our findings could show that the certification potentially exacerbated the effect of system reliability. However, without a measure of the appropriateness of reliance, it remains unclear whether the certification made participants overly rely on the system or to the right amount.

Additionally, while we focus on certifications as a key factor in trust, the study does not delve deeply into the underlying psychological mechanisms that influence how certifications are perceived. Factors such as cognitive load, ease of use, and familiarity with the technology may play crucial roles in shaping user responses to certifications. For example, users with less experience with AI may rely more heavily on certifications as a signal of reliability, while more experienced users may approach certifications with a more critical eye. Understanding how these factors interact with certifications to affect trust would provide a richer picture of the dynamics at play. Future work could examine how cognitive load and other individual differences impact users' perceptions and decisions about certified AI systems.

Another limitation of our study is its focus on reliability as a central factor in building trust. While reliability is undoubtedly important, other factors like fairness, explainability, and transparency are also key drivers of trust in AI systems. Recent research, such as Cetinkaya and Krämer [18], has shown that different users prioritize different attributes in AI systems. Some users may place more value on system fairness, while others may prioritize explainability or the mere existence of a certification. We speculate that other individual differences, such as, for example, AI expertise might also influence the effectiveness of the certifications. Users who are more experienced with AI might rely less on certifications but, instead, prefer more direct benchmarks of systems. However, as we were interested in how the general population perceives and interacts with certified AI systems, we did not include levels of expertise in our study.

Moreover, the limited value of certifications after interaction presents another challenge. Once an AI system has been certified, that certification often remains static unless a new evaluation is conducted. However, as users engage with the system over time, their trust and expectations evolve based on the system's actual performance. Our study found that the impact of certification tends to diminish after users begin interacting with the system, likely because certifications alone do not account for dynamic, real-world system behavior. Future studies could explore how certifications can be updated or complemented with ongoing monitoring and performance evaluations to ensure they remain relevant throughout the lifecycle of the AI system.

Finally, it is important to note that the voluntary nature of the certification in this study may have influenced the results. If certification were mandatory for all AI systems, the effects might have differed. Users could potentially view mandatory certification as a regulatory obligation, which might reduce the perceived value of the certification or alter the way they interact with the system. Surely, we could expect that, if certifications were mandatory, any system that is not certified would not be trusted. However, this should be empirically explored.

6 Conclusion

In this work, we examined how AI certifications interact with system reliability to shape users' trust, vigilance, reliance, and expectation formation. Our findings demonstrate that certifications function primarily as initial trust cues, boosting pre-interaction trust and lowering vigilance. However, these effects diminish once users begin interacting with the system, where reliability significantly affected trust formation. Certifications nonetheless continued to influence reliance behaviors, suggesting that institutional signals can persist as heuristics for decision-making even when experiential evidence contradicts them. Additional exploratory results showed that the certification supported appropriate reliance, especially when a system was less reliable and made more errors. We surmise that, for those cases, the certification protected participants against the overgeneralization of the AI errors.

Importantly, however, our study reveals a double-edged nature of certifications. When paired with unreliable system performance, certifications amplified negative expectation violations and, through this, reduced trust, while simultaneously supporting appropriate reliance. Conversely, when reliability was high, certifications reinforced trust. These results underline the fragility of certifications as governance tools: they can raise confidence at the outset, but also risk undermining trust calibration when performance does not meet user expectations.

Taken together, our findings contribute to ongoing debates about the design and implementation of AI governance mechanisms. For policymakers, they suggest that certifications should be accompanied by clear communication of scope and limitations, as well as mechanisms for ongoing monitoring and renewal rather than one-time assessments. For HCI, our results emphasize the need to examine not only whether certifications increase trust, but also whether they support appropriate reliance over time.

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